LIN7209 – Syntax

Extended projections

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07/10/2025

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Plan for today

- Review of phrase structure, types of heads, θ-roles.
- Extended projections: description and advantages.

Heads and structural relations

- Constituents tend to inherit the distribution and properties of one of their elements–the **head**.
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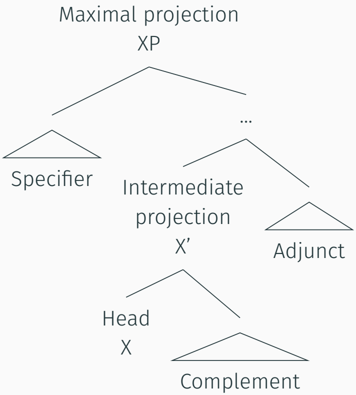

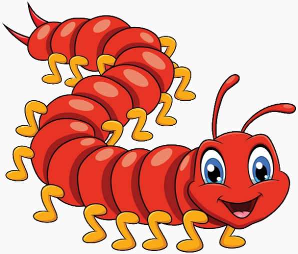

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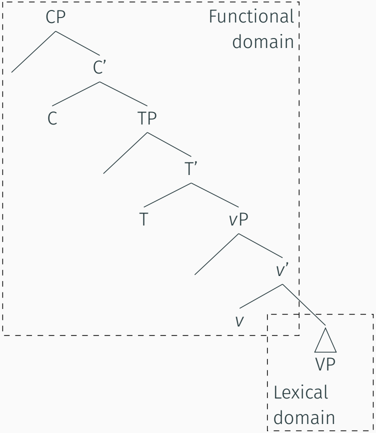
Functional vs. lexical



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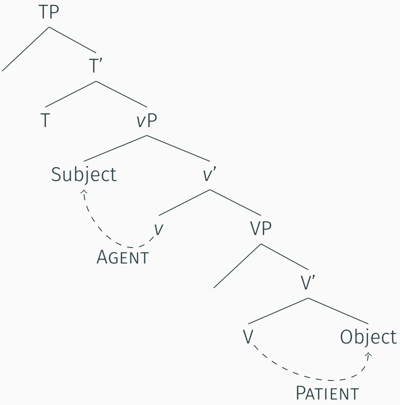
- **Functional** heads are closed-class, phonologically weak, semantically fixed, "functional"/"relational" elements. Examples: complementizers, tense, light verbs (DO, CAUSE...).
- At the clausal level, functional heads tend to be **higher** than lexical ones.

The verbal domain

- The complement (object) of the lexical verb head (V) is its only true argument. This **object** often gets a PATIENT Θ-role.
- *v* is a functional head which hosts a "light" verb (CAUSE, DO), covert in English. *v* assigns a Θ-role to its specifier (the **subject**), often AGENT.



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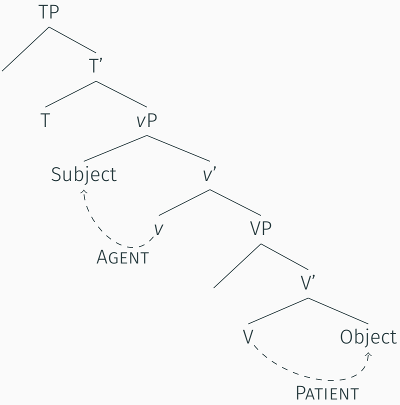

Morphological evidence for *v*

- Malagasy (Austronesian), Hiaki (Uto-Aztecan), Japanese (Japonic) have an affix for *v* meaning “active voice”/CAUSE.

(1) Malagasy:

M-**an**-sasa	ny	lamba	amin	ny	savony	Rasoa
PST-**v**-wash	the	clothes	with	the	soap	Rasoa

‘Rasoa causes the clothes to be clean (=washes) with the soap.’

(2) Hiaki:

Huan	u’usit-ta	ee-**tua**-k.
Juan	child-ACC	feel-**v**-PST

‘Juan teased the child.’

(3) Japanese:

Keiko-wa	pizza-o	ag-**e**-ta
Keiko-TOP	pizza-ACC	rise-**v**-PST

‘Keiko raised the pizza.’

Semantic evidence for *v*

- If a given constituent is a true argument of a head, then the head may be sensitive to its semantic characteristics.
- The interpretation of V is often sensitive to the semantic characteristic of its object, but never to those of its subject!

(4) a. **throw** a baseball. throw ≃ cast
b. **throw** support behind a candidate. throw ≃ provide
c. **throw** a party. throw ≃ organize

- This implies subjects are not true arguments of the verb, and motivates a structure whereby subjects are not specifiers of V, but instead specifiers of a higher **functional head**, namely *v*.

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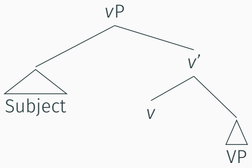

Testing for AGENCY

- Purpose clauses.

(5) Jo smiled in order to reassure Ed.
  AGENT

(6) ?? Jo heard the conversation to gain new insights.
  EXPERIENCER

(7) ?? Jo fell in order to distract the police.
  PATIENT/THEME

- Adverbs like *deliberately*, *intentionally*

(8) Jo {smiled / ??heard the conversation / ??fell } deliberately.

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More thematic roles (Theme is a little bit of a catch-all)

(9) a. Jo (Patient/Theme) fell. — Unaccusative

b. Jo (Agent) jumped. — Unergative

(10) a. Jo (Agent) melted the ice. (Theme) — Causative

b. The ice (Theme) melted. — Inchoative

(11) a. Jo (Experiencer) fears snakes. (Stimulus/Theme) — Psych verb

b. Snakes (Stimulus/Theme) frighten Jo. (Experiencer) — Psych verb

(12) a. The book (Theme) is blue. — Copular/Predicative construction

b. The ball (Theme) rolled away. — Change-of-location

c. Jo (Agent) gave Ed (Goal) the book (Theme). — Ditransitive

- Which role(s) get(s) assigned depends on the semantics of the verb, defining its "Θ-grid".

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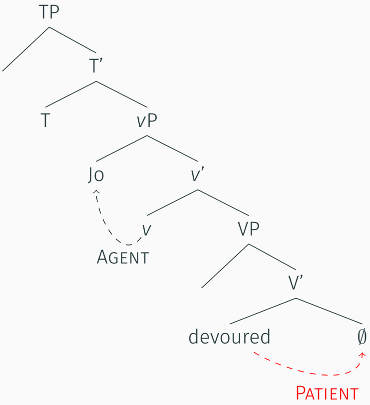

- If a strongly transitive verb does not have an object, a Θ-role is unassigned, and the Θ-grid of the verb is not "exhausted". Ungrammaticality arises.

Too many arguments

(14) Jo arrived *(an arrival). Unaccusative

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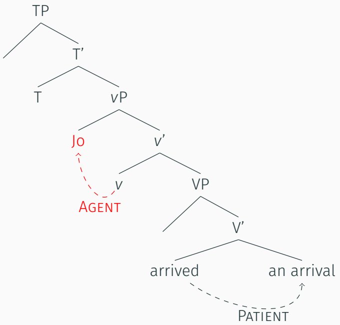


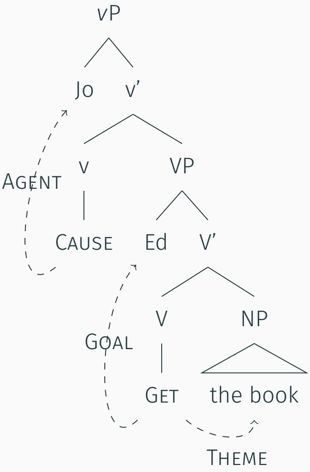
- If the *v* in unaccusative structures assigned an AGENT Θ-role, then we could not explain why such structures are happy with one single PATIENT argument (see previous slide).
- So, *v* must be defective (empty Θ-grid), and then *Jo* gets no Θ-role... causing ungrammaticality.

Constraints on Θ-roles

- The **Θ-criterion**:[1] each Θ-role gets assigned to one and only one argument, and each argument receives one and only one Θ-role.
- The **Uniformity of Theta-Assignment Hypothesis** (UTAH):[2] Identical **thematic** relationships between items are represented by identical **structural** relationships between those items at deep structure (i.e. before movement/Internal Merge).
    - Agent = causer = Spec-*v*P
    - Patient = undergoer = Comp-V
- Θ-roles are assigned **locally**, typically in a Comp-Head or Spec-Head configuration.

[1] Chomsky, 1981.
[2] Baker, 1988, 1997.

Ditransitives (Adger, 2003)



(15) Jo gave Ed the book.

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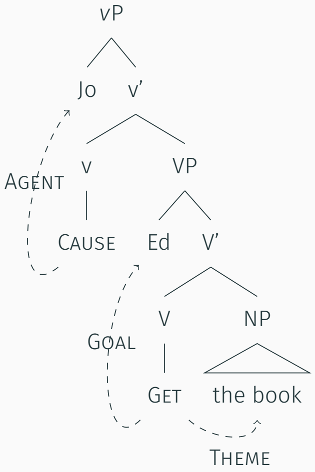

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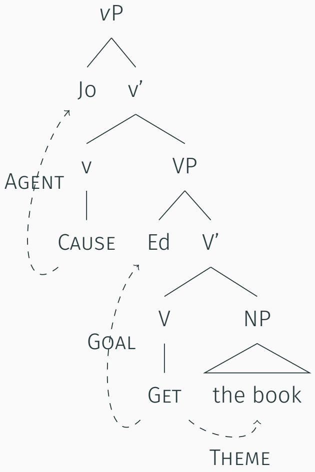

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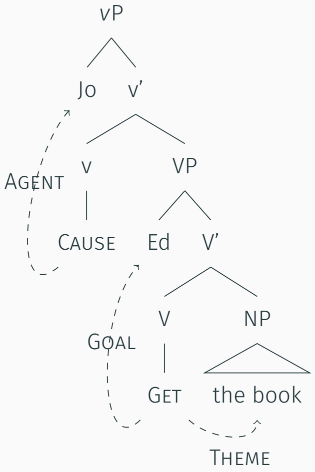

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Extended projections

Comparing verbs and nouns

- Verbs and nouns assign very similar Θ-roles.

(16)
a. $\underset{\text{AGENT}}{\underline{\text{Jo}}}$ ’s **interview** of $\underset{\text{PATIENT/THEME}}{\underline{\text{the candidate}}}$
b. $\underset{\text{EXPERIENCER}}{\underline{\text{Jo}}}$ ’s **love** of $\underset{\text{STIMULUS/THEME}}{\underline{\text{music}}}$
c. $\underset{\text{Kin}}{\underline{\text{Jo}}}$’s **mother**.
d. A **cup** of tea, the **top** of the desk, the **King** of France, **paintings** of Jo...

- Both verbs and nouns generally come with a bunch of other projections above them: CP, TP, *v*P for verbs, PP, DP for nouns.
- A lot of symmetries between verbs and nouns! How to cash this out?

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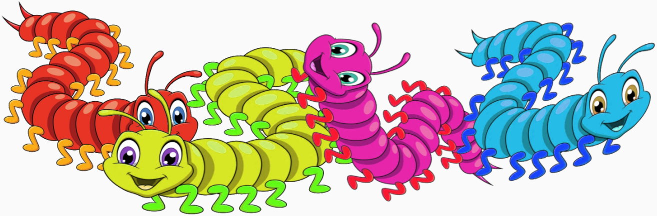
Assigning structure to structure

- Idea: we don't have isolated projections (the "centipedes"), we have a bunch of them having a big party together.
- Groups of related projections are called **extended projections**.[3]
- Main advantage: explain why relations between projection that are expected to be **local** (between adjacent projections), sometimes appear **non-local** (i.e. seemingly "skip" projections).



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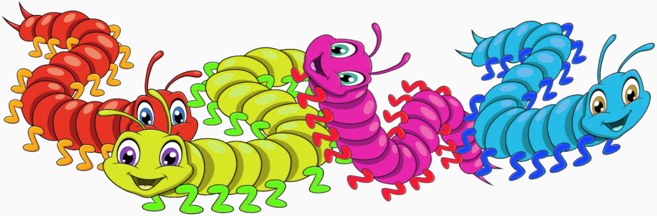

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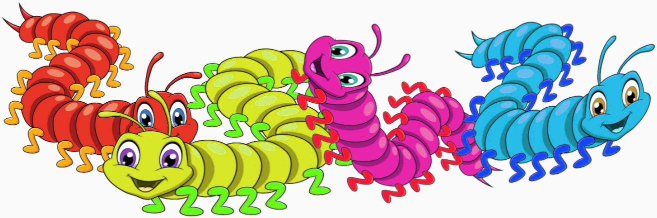

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- All projections (functional/lexical) belong to the same overall "**category**"/**domain**: typically verbal or nominal.
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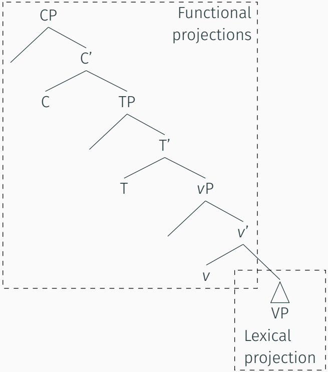

- V is a lexical head.
- *v*P, TP, CP are functional projections stacked over it.
- $\{CP, TP, vP, VP\}$ forms an **extended projection of the verbal category**.
- In this extended projection, the lexical projection VP is downstairs, the functional ones upstairs.

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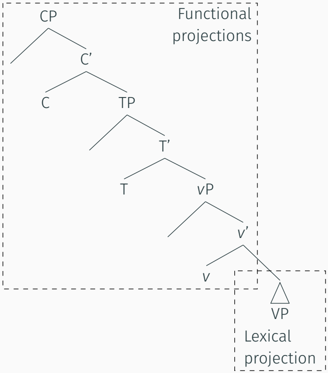

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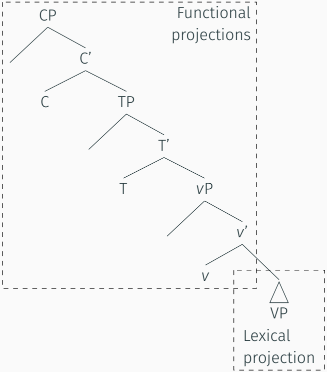

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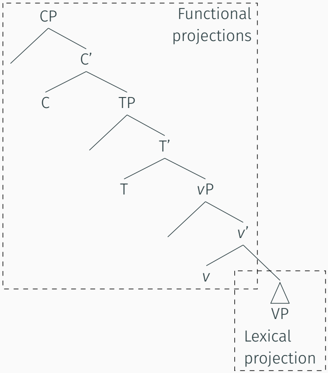

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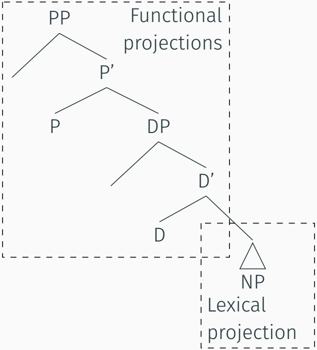

- N is a lexical head.
- DP, PP are functional projections that can be stacked over it.
- $\{PP, DP, NP\}$ forms an **extended projection of the nominal category**.
- In this extended projection, the lexical projection NP is downstairs, the functional ones upstairs.
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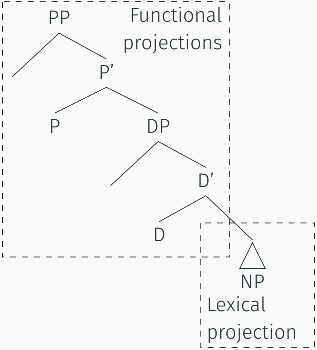

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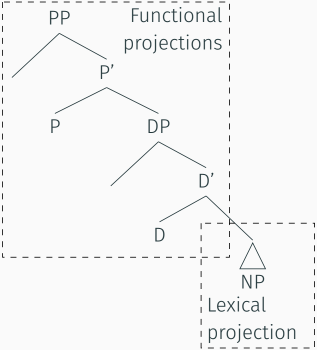

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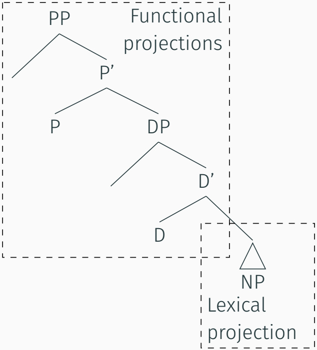

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Arguments for an extended projection in the nominal domain

V-heads can access N-features across the D projection

- Selection: verbs can select certain features on the noun, **even in the presence of an intervening determiner** forming its own projection!

(17) They merged [$_{DP}$ the [$_{NP}$ files / *file / dogs]].

- Agreement: sensitive to the features of the head, again despite intervening determiners!

(18) [$_{DP}$ The [$_{NP_{[PL]}}$ dogs]] {is/*are}...

(19) [$_{DP}$ [$_{DP}$ The boy] [$_{D}$'s] [$_{NP_{[SG]}}$ book]] {is/*are}...

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- *Wh*-movement: we expect the *wh*-phrase (a DP) to be fronted.
- However, its also possible to front the entire PP, whose complement is the *wh*-phrase! This is called **pied-piping**.
- It's also active in English relative clauses.

(20) a. They found a note [$_{\text{PP}}$ under [$_{\text{DP}}$ **that stone**]].

b. [$_{\text{DP}_{[+wh]}}$ **Which stone**] did they find a note [$_{\text{PP}}$ under __]?

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- Extra evidence for the fact that PP belongs to N's extended projection comes from the selection of interrogative complements by verbs like *wonder*.
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Arguments for an extended projection in the verbal domain

Mood selection

- Certain verbs selecting for the same complementizer, **force distinct moods** on the embedded verb...
- This is surprising under a purely local account: we'd not expect the embedding V to mess up with the embedded T (hosting MOOD) across C, plus **we'd expect the same Cs to select for the same Ts**!

(22) a. Jo requested [$_{CP}$ **that** [$_{TP}$ Ed {leave.SBJV/?left.IND} at 6]].
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